CHEMICAL EQUILIBRIUM 27

Generally changes of state are **reversible**; for example, liquid water can be changed to solid ice and vice versa. Most chemical reactions are theoretically reversible. This means that not only do reactants form products; but (depending on conditions) products can be changed back to the original reactants. Equilibrium reactions are special types of reversible reactions.

Alexander William Williamson (1824–1904) was the first person to systematically study reversible reactions. He was also the first person to come up with the concept of **dynamic equilibrium**. Dynamic equilibrium describes the equilibrium, or balance, of forward and reverse reactions happening simultaneously. Therefore an equilibrium reaction involves two reactions that are occurring at the same time but in opposite directions. In other words, reactants are being changed to products at the same time that products are being changed to reactants. At equilibrium, the rates of the forward reactions and reverse reactions are the same. This is shown by using the equilibrium symbol ($\rightleftharpoons$).

For example: Products $\rightleftharpoons$ Reactants

1 Write in words the equation for each reaction.

a $CaCO_3$ (s) (heat, open system) $\rightarrow$ CaO (s) + CO_2 (g)

b CaO (s) + CO_2 (g) (pressure, closed system) $\rightarrow$ $CaCO_3$ (s)

Some reactions can be reversed by changing reaction conditions. For example, if calcium carbonate is heated in a closed vessel, at first the only reaction will be calcium carbonate decomposing to form calcium oxide and carbon dioxide. When the pressure and the concentration of calcium oxide and carbon dioxide reach a certain point, they start to react to form calcium carbonate. This is shown by the equation: $CaCO_3$ (s) $\rightleftharpoons$ CaO (s) + CO_2 (g)

2 Will an equilibrium reaction in a sealed system ever reach completion, i.e. give 100% products? Explain your answer.

Summary

- Reactants combine to make products. (The forward reaction.)
- In some reactions, products can break apart and turn back into the reactants. (The reverse reaction.)
- At equilibrium, forward and reverse reactions are taking place at the same rate.
- At equilibrium it *seems* as if no reactions are occurring.
- Equilibrium reactions do not go to completion; at equilibrium there is a mixture of reactants and products.

When talking about equilibrium reactions, the terms **micro** and **macro** are often used. At equilibrium, there are reactions taking place that are only visible at the microscopic level. At the larger (macro) level, the naked eye can't see any change.

3 Suggest why it seems that no reactions are occurring at equilibrium.

4 Define *micro* and *macro*.

5 Study the following description of an equilibrium reaction and sketch a graph to show how this reaction proceeds. Graph concentration on the y-axis and time on the x-axis.

In an equilibrium reaction, at the moment the reactants are mixed the concentration of reactants is 100% and the concentration of products is 0%. As time passes, the concentration of reactants decreases, reaches a minimum point and stays at that concentration. At the same time the concentration of products increases, reaches a maximum point and stays at that concentration.

6 Copy this statement and include the missing word:

For a chemical reaction at _____, the amount of reactants being transformed to products per unit time is exactly matched by the amount being transformed to reactants (from products) per unit time.

7 *The nature and properties of the equilibrium state remain the same, regardless of the direction from which it is reached.* Explain what is meant by this statement.

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Catalysts and equilibrium reactions

Figure 1 shows the reaction profile for an exothermic equilibrium reaction. You can see that the activation energy (E_a) for the catalysed reaction is less than the activation energy for the uncatalysed reaction. Catalysts speed up both the forward and reverse reactions equally. This means that adding a catalyst to an equilibrium reaction causes that reaction to reach equilibrium more quickly, but does not alter anything else.

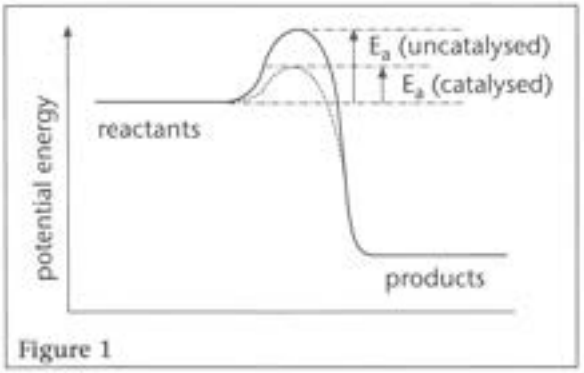


Figure 1

Changing equilibrium systems

The position of equilibrium varies from reaction to reaction. Often chemists try to affect equilibrium reactions. Reasons for this include speeding up the reaction and increasing the yield of products.

8 Name three things a chemist could do to affect an equilibrium reaction.

Henri-Louis Le Châtelier (1850–1936) made great contributions to our understanding of equilibrium systems. He is famous for **Le Châtelier's principle**. One version of his principle is: *A system in equilibrium, when subjected to a stress resulting from a change in temperature, pressure or concentration, and causing the equilibrium to be upset, will adjust its position of equilibrium to relieve the stress and re-establish equilibrium.*

This can be summarised as follows.

- An equilibrium system will not change its rate of forward and reverse reactions until conditions are changed.
- Equilibrium reactions can be affected by changes in concentration, heat and (when gases are involved) pressure.
- An equilibrium reaction will change in order to reduce the stress on the system.

It doesn't matter how the conditions of the system change, there are only two possible outcomes:

- The forward reaction speeds up to produce more products
- The reverse reaction speeds up to produce more reactants.

So when conditions change, the forward/reverse reactions change; after some time, the reaction will reach a **new** equilibrium point.

Changes in concentration

If the concentration changes in the product or the reactant, the forward and reverse reactions will alter to minimise this change.

EXAMPLE

i **Look at the reaction between hydrogen and iodine forming hydrogen iodide:**
H_2 (g) + I_2 (g) ⇌ 2HI (g)

If hydrogen is added to the above reaction at equilibrium point, the forward reaction will increase in speed (in order to remove 'excess' hydrogen and produce more hydrogen iodide) until a new equilibrium is reached.

If hydrogen iodide is added at equilibrium point, the reverse reaction will speed up (in order to remove 'excess' hydrogen iodide and produce more hydrogen and iodine) until a new equilibrium is reached.

9 Use Le Châtelier's principle to predict what would happen to the above equilibrium system if hydrogen iodide was removed.

10 What would happen to the reaction between hydrogen and iodine forming hydrogen iodide if a catalyst was used? Explain your answer.

Esterification is an equilibrium process. One example is when ethanol reacts with ethanoic (acetic) acid to form ethyl ethanoate (an ester) and water.

$$C_2H_5OH\ (l) + CH_3COOH\ (l) \rightleftharpoons CH_3COOC_2H_5\ (l) + H_2O\ (l)$$

11 Outline two ways that the yield of ester could be increased.

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Changes in heat

Reactions that give out heat are exothermic (i.e. ΔH is negative). Reactions that take in heat are endothermic (i.e. ΔH is positive). Heating or cooling equilibrium reactions will have different effects, depending on whether these reactions are exothermic or endothermic. Consider heat as a reactant or product in reactions (not strictly true, but a useful simplification). This can help in using Le Châtelier's principal to predict changes due to heat. A reaction that is exothermic going from reactants to products (forward reaction) will be endothermic going from products to reactants (reverse reaction). If heat is a 'product' then adding more heat will cause the reaction to try to remove this heat.

- Adding heat to an exothermic reaction (that 'wants' to lose heat) will oppose the forward reaction and speed up the reverse (endothermic) reaction.
- Adding heat to an endothermic reaction (that 'wants' to gain heat) will speed up the forward reaction and oppose the reverse (exothermic) reaction.

12 Copy each reaction and use Le Châtelier's principle to predict what would happen if the temperatures were changed from standard conditions.

a	$2NO_2\ (g) \rightleftharpoons N_2O_4\ (g)$	$\Delta H = -58.1$ kJ	(heated)
b	$C\ (s) + H_2O\ (g) \rightleftharpoons CO\ (g) + H_2\ (g)$	$\Delta H = +131$ kJ	(heated)
c	$2SO_2\ (g) + O_2\ (g) \rightleftharpoons 2SO_3\ (g)$	$\Delta H = -197$ kJ	(cooled)
d	$N_2\ (g) + O_2\ (g) \rightleftharpoons 2NO\ (g)$	$\Delta H = +182$ kJ	(cooled)

Changes in volume

Changes in pressure can influence equilibrium reactions where gases are present. However, there also needs to be a difference in the number of moles of products compared to reactants. If pressure is changed, Le Châtelier's principle says that the system will react to try to restore original pressure. Increasing pressure causes a decrease in the volume of a gas; decreasing pressure causes an increase in the volume of a gas (assuming temperature remains constant).

- An increase in pressure will favour the reaction that decreases volume.
- A decrease in pressure will favour the reaction that increases volume.
- Solids and liquids present are assumed to take up minimal volume.

EXAMPLE

ii Study this reaction: $2SO_2\ (g) + O_2\ (g) \rightleftharpoons 2SO_3\ (g)$

The reactants take up 3 volumes (of a given unit) and the products take up 2 volumes. In other words a volume ratio of 3:2.

Therefore an increase in pressure would favour a decrease in volume, driving the equilibrium system from left to right.

13 Copy each reaction and use Le Châtelier's principle to predict what would happen if the pressure was changed from standard conditions.

a	$2NO_2\ (g) \rightleftharpoons N_2O_4\ (g)$	(pressure increased)
b	$C\ (s) + H_2O\ (g) \rightleftharpoons CO\ (g) + H_2\ (g)$	(pressure increased)
c	$2SO_2\ (g) + O_2\ (g) \rightleftharpoons 2SO_3\ (g)$	(pressure decreased)
d	$N_2\ (g) + O_2\ (g) \rightleftharpoons 2NO\ (g)$	(pressure decreased)

14 Summarise Le Châtelier's principle.

15 What is a *prediction*?

Research **16** In science you will hear about *Theories* and *Principals* or *Laws*. Theories are different from Principals/Laws. Find out how they are different.

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